House prices analysis

1. Project Overview

This project analyzes a house prices dataset to understand the factors that influence property prices. The main goal is to explore the dataset, clean and validate the data, perform statistical analysis, and visualize patterns that affect housing prices.

The analysis uses Python libraries such as **pandas, NumPy, matplotlib, and seaborn** to manipulate data and generate visual insights. By examining variables like number of bedrooms, bathrooms, and living area, the project aims to identify relationships and trends that can help in understanding real estate price patterns.

The outcome of this analysis includes data visualizations, statistical insights, and observations that could help buyers, sellers, or real estate analysts make informed decisions.

2. Setup Instructions

Follow these steps to run the project:

1. Install **Python (version 3.8 or higher)** on your system.
2. Install **Anaconda or Jupyter Notebook** for running the notebook environment.
3. Install the required Python libraries using the following command:

pip install pandas numpy matplotlib seaborn jupyter

4. Download the dataset file **house_prices.csv**.
5. Place the dataset in the same folder as the Jupyter Notebook.
6. Open **Jupyter Notebook** and navigate to the project folder.
7. Run the notebook cells sequentially to perform the data analysis.

3. Code Structure

The project code is organized into multiple sections to ensure clarity and maintainability.

1. Import Libraries
This section imports required libraries such as pandas, NumPy, matplotlib, and seaborn.

2. Data Loading
The dataset is loaded using pandas and the first few rows are displayed to understand the structure.

3. Data Exploration
Initial exploration is performed to understand dataset shape, column names, and data types.

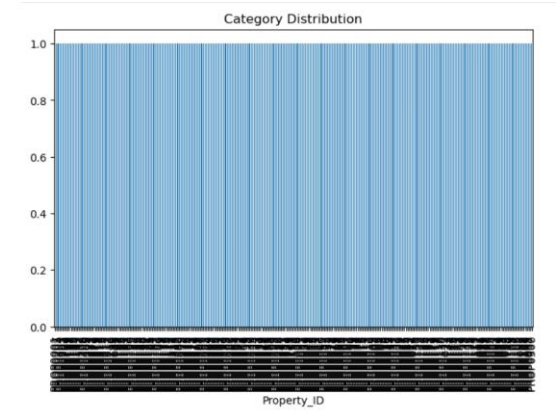
4. Data Cleaning and Validation
Missing values are handled, duplicate records are removed, and data types are corrected if necessary.

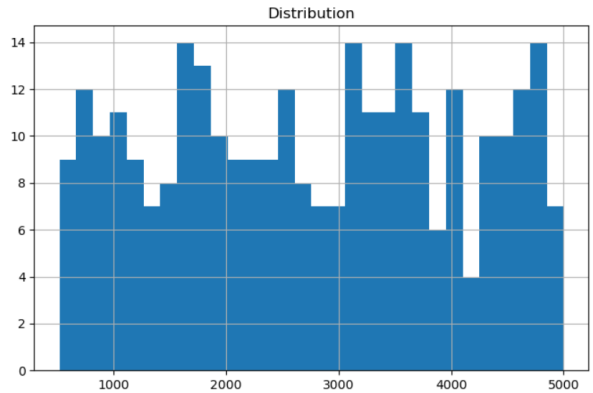
5. Statistical Analysis
Basic statistical measures such as mean, median, and correlations are calculated.

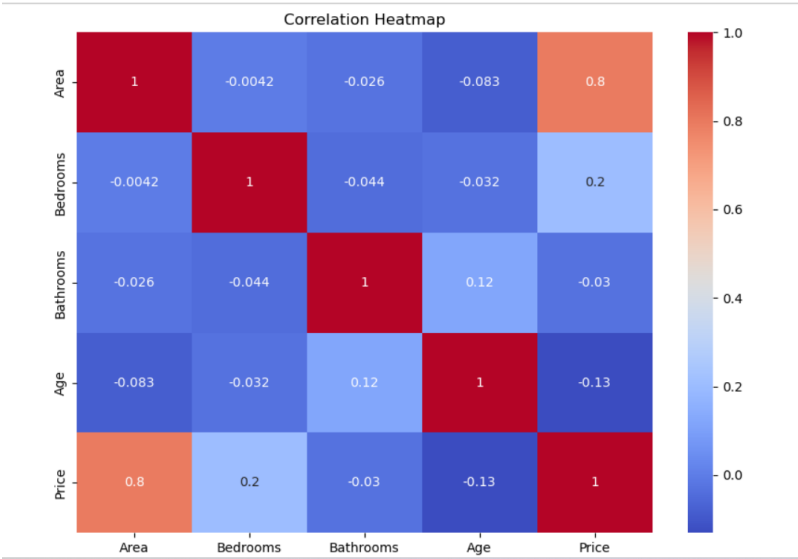
6. Data Visualization
Several visualizations such as histograms, bar charts, and correlation heatmaps are created to understand patterns in the data.

7. Insights and Observations
Key findings from the analysis are summarized to highlight important trends in the housing data.

4. Visualization







5. Technical Details

This project uses Python-based data analysis techniques to process and analyze the house prices dataset. The main libraries used are pandas, NumPy, matplotlib, and seaborn.

Algorithms and Methods

The analysis uses several basic statistical and data processing methods:

- Mean and Median Calculation – These statistical measures help understand the average and central tendency of house prices and other numerical features.

- Correlation Analysis – The correlation matrix is used to measure the relationship between variables such as price, number of bedrooms, bathrooms, and square footage.
- Grouping and Aggregation – The groupby() function in pandas is used to analyze trends across categories such as number of bedrooms or floors.
- Data Visualization Techniques – Histograms, bar charts, and heatmaps are used to visually represent the distribution and relationships within the data.

Data Structures

The project mainly uses pandas DataFrames, which are two-dimensional data structures used to store and manipulate tabular data. Each column represents a variable and each row represents a record in the dataset.

Lists and arrays from NumPy are also used internally for numerical computations and statistical operations.

Architecture

The project follows a simple data analysis pipeline architecture:

1. Data Collection – Import the dataset from a CSV file.
2. Data Preprocessing – Clean the data by handling missing values and removing duplicates.
3. Data Analysis – Apply statistical methods to understand patterns in the data.
4. Visualization – Generate visual graphs to represent the findings.
5. Insights Generation – Interpret the results and derive meaningful conclusions.

This modular structure makes the analysis easy to understand, maintain, and extend.

6. Testing Evidence

Testing was performed to ensure that the dataset was correctly processed and that the analysis results were accurate.

Data Validation Tests

Several validation checks were applied to verify data quality:

- Missing Value Check – Verified using df.isnull().sum() to ensure no critical data fields were missing.
- Duplicate Data Check – Used df.duplicated().sum() to identify and remove duplicate records.
- Data Type Verification – Checked column data types using df.info() to confirm correct formats.

Output Verification

The correctness of the analysis was validated through:

- Reviewing statistical outputs such as mean, median, and correlation values.
- Confirming that visualizations (histograms, bar charts, heatmaps) accurately represented the dataset.
- Ensuring that the cleaned dataset produced consistent results across multiple runs.

Functional Testing

Each stage of the notebook was executed sequentially in Jupyter Notebook to confirm that:

- The dataset loads correctly.

- Data cleaning steps execute without errors.
- Visualizations are generated successfully.
  These tests confirm that the data analysis workflow functions correctly and produces reliable results.